

Trader Behaviour under Bitcoin Fear & Greed Sentiment

1. Objective

- Understand how **trader performance** (PnL, win rate) changes with **market sentiment** (Fear & Greed Index).
- See how **trading behaviour** (number of trades, position size, fees) differs between **Extreme Fear, Fear, Neutral, Greed, Extreme Greed**.
- Compare **top-performing** and **bottom-performing** traders under these sentiment regimes.
- Look at **trends over time**: how sentiment, activity, and PnL move together.

2. Data & Features

- Historical trade data aggregated to **account-day** level.
- Engineered metrics per account-day:
 - total_daily_pnl: net PnL for the day.
 - win_rate: share of winning trades.
 - num_trades: number of trades.
 - avg_size_usd: average trade size in USD.
 - avg_fee_usd: average fee per trade.
- Joined with **daily Fear & Greed Index (value)** and a **sentiment_class** label: Extreme Fear, Fear, Neutral, Greed, and Extreme Greed.

	Account	trade_date	total_daily_pnl	num_trades	wins	avg_size_used	avg_fee	win_rate	value	sentiment_class
0	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-11	0.000000	177	0	5089.718249	0.948000	0.000000	76.0	Extreme Greed
1	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-17	0.000000	68	0	7976.664412	0.998288	0.000000	90.0	Extreme Greed
2	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-18	0.000000	40	0	23734.500000	2.373450	0.000000	83.0	Extreme Greed
3	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-22	-21227.000000	12	0	28186.666667	2.818666	0.000000	94.0	Extreme Greed
4	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-26	1603.100000	27	12	17248.148148	3.874907	0.444444	79.0	Extreme Greed
...	...	...	...	...	...	...	...	...	...	...
2336	0xbec1707d6b44d4d52bfe19e41f8a828645437aab	2025-04-27	1709.194807	82	32	2153.859390	0.312681	0.390244	61.0	Greed
2337	0xbec1707d6b44d4d52bfe19e41f8a828645437aab	2025-04-28	4008.588908	430	169	1495.538419	0.371047	0.393023	54.0	Neutral
2338	0xbec1707d6b44d4d52bfe19e41f8a828645437aab	2025-04-29	8561.771838	902	398	1939.739989	0.386831	0.441242	60.0	Greed
2339	0xbec1707d6b44d4d52bfe19e41f8a828645437aab	2025-04-30	2520.773814	75	39	1748.805333	0.270870	0.520000	56.0	Greed
2340	0xbec1707d6b44d4d52bfe19e41f8a828645437aab	2025-05-01	1364.022527	131	43	2448.466489	0.457253	0.328244	53.0	Neutral

2341 rows x 10 columns

3. Methods (very brief)

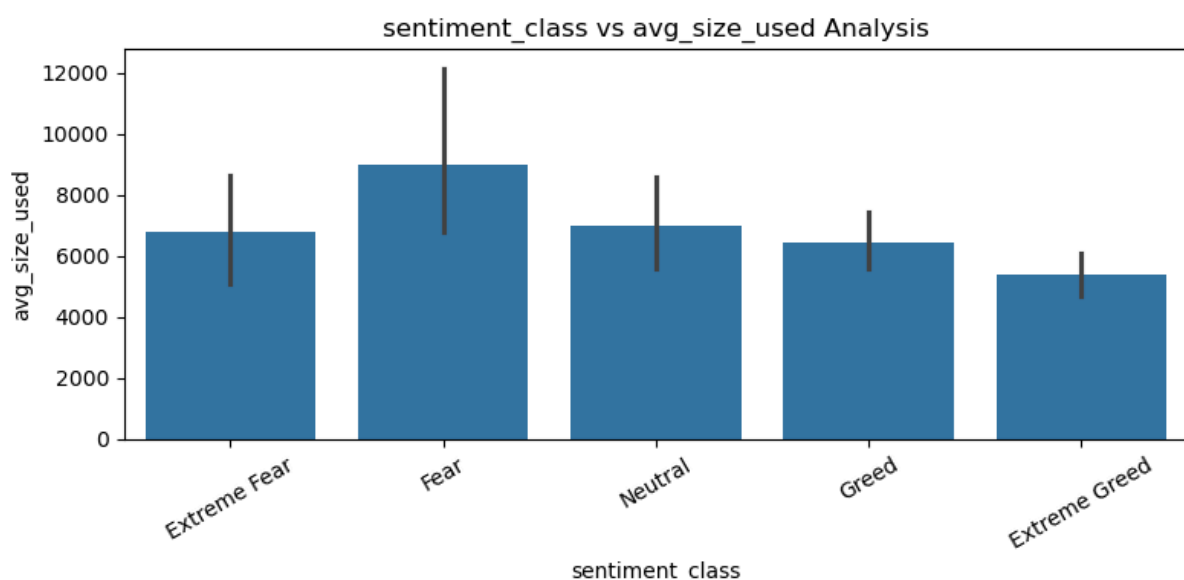
- Used **Python (pandas, Seaborn)** in notebook_2.
- Grouped data by **sentiment_class** and by **Account + sentiment_class**.
- Created **Top** and **Bottom** trader groups based on **average daily PnL**.
- Calculated **correlation** between numeric features.
- Built **daily time-series** and used a **7-day rolling average** to smooth PnL, sentiment, and number of trades.

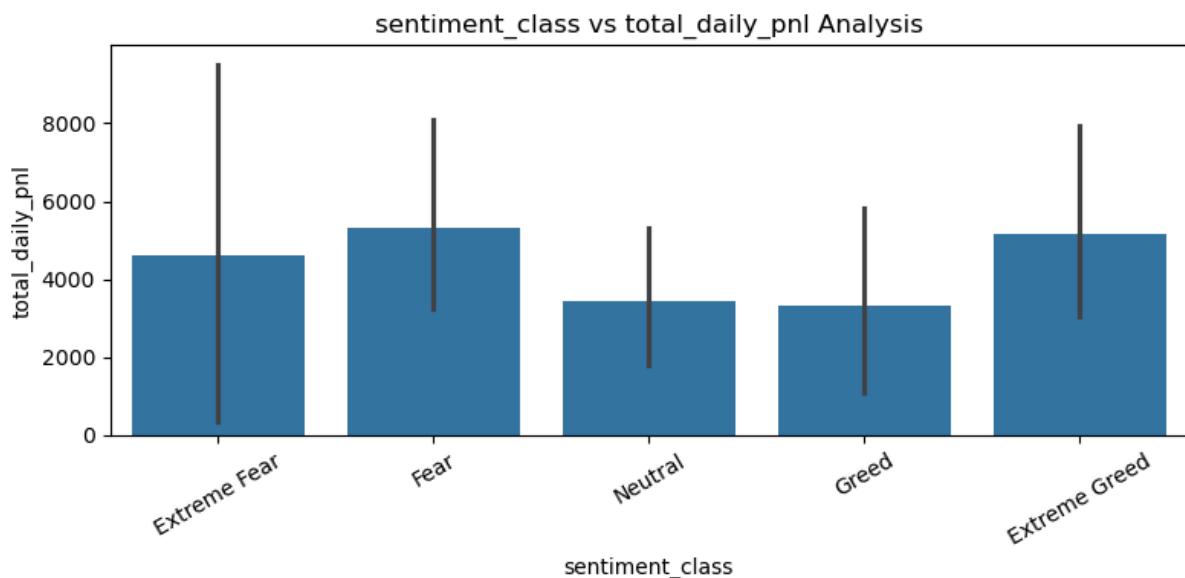
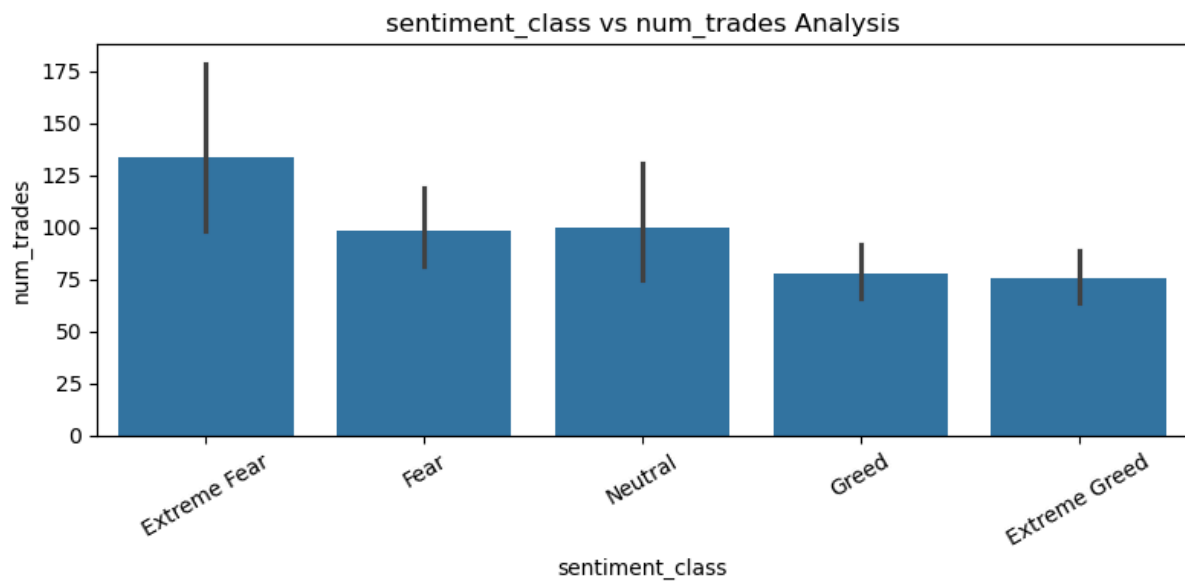
5. Key Insights

1. Market sentiment changes how traders behave

The data shows that trader activity and risk-taking vary significantly across sentiment regimes.

During Extreme Fear, traders place more trades and show highly volatile PnL and win rates, indicating panic-driven and unstable behaviour. In contrast, during Extreme Greed, traders execute fewer trades with smaller position sizes and lower fees, yet achieve higher win rates and more consistent profits. This suggests that more disciplined and selective trading occurs when the market is optimistic rather than fearful.





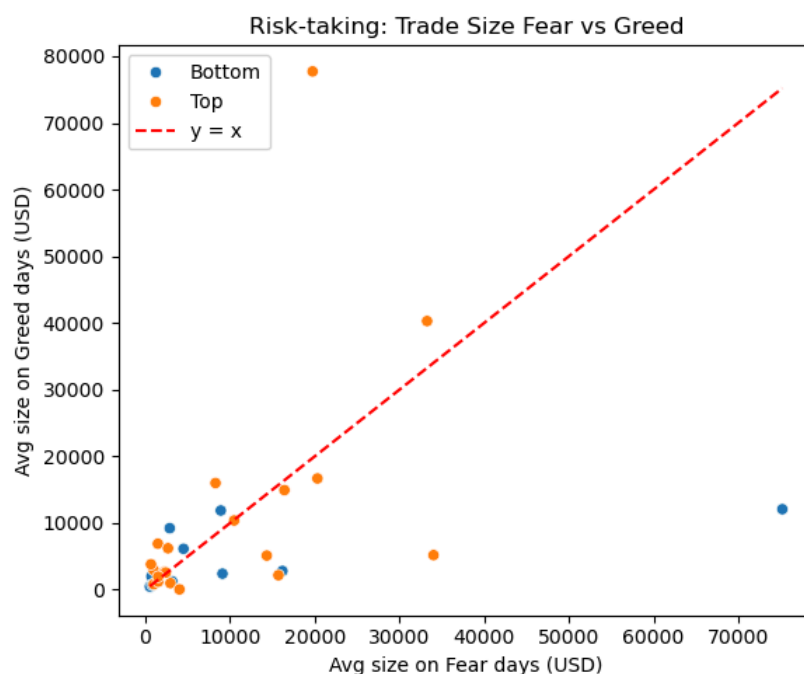
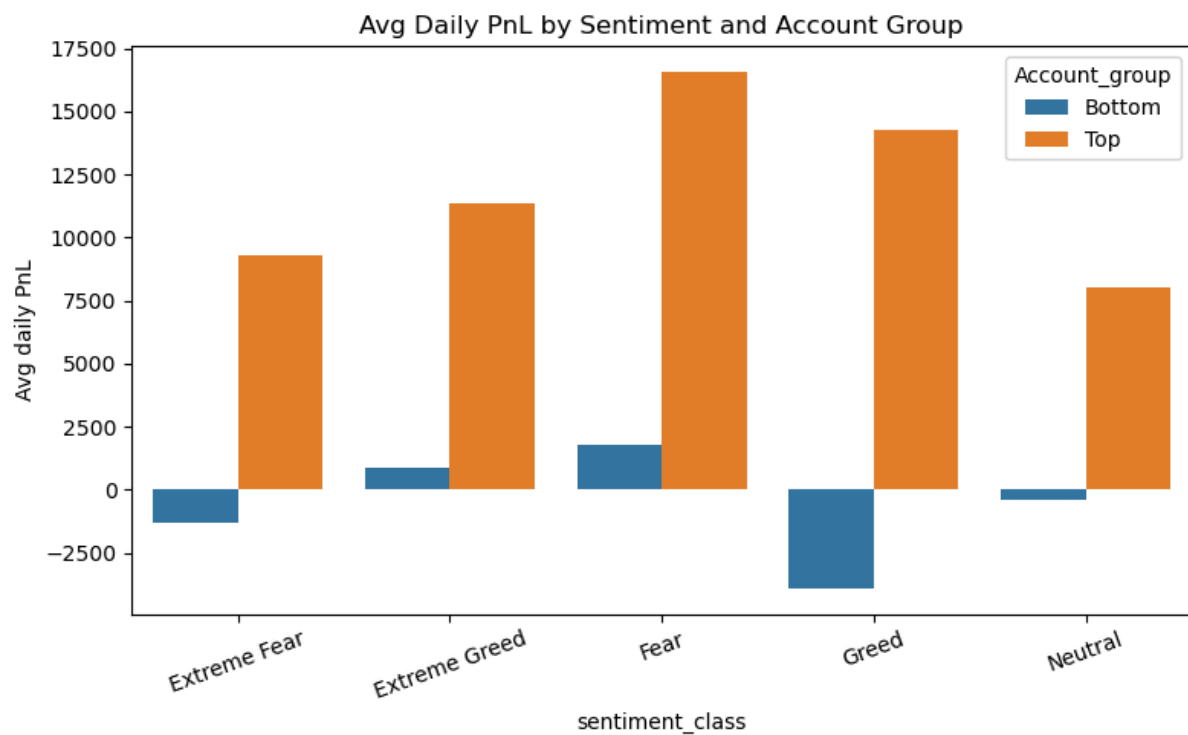
2 Top traders outperform in every market condition

When accounts are grouped by average daily PnL, a clear separation emerges between top and bottom traders.

Top traders remain profitable across all sentiment regimes, including during Fear and Extreme Fear periods. Bottom traders, however consistently lose money even when market sentiment is Greedy or Extreme Greedy. This implies that skills and strategy dominate market sentiment in determining long-term profitability.

Furthermore, top traders tend to increase their position size and sometimes trade frequency during Greed phase, while bottom traders maintain small and

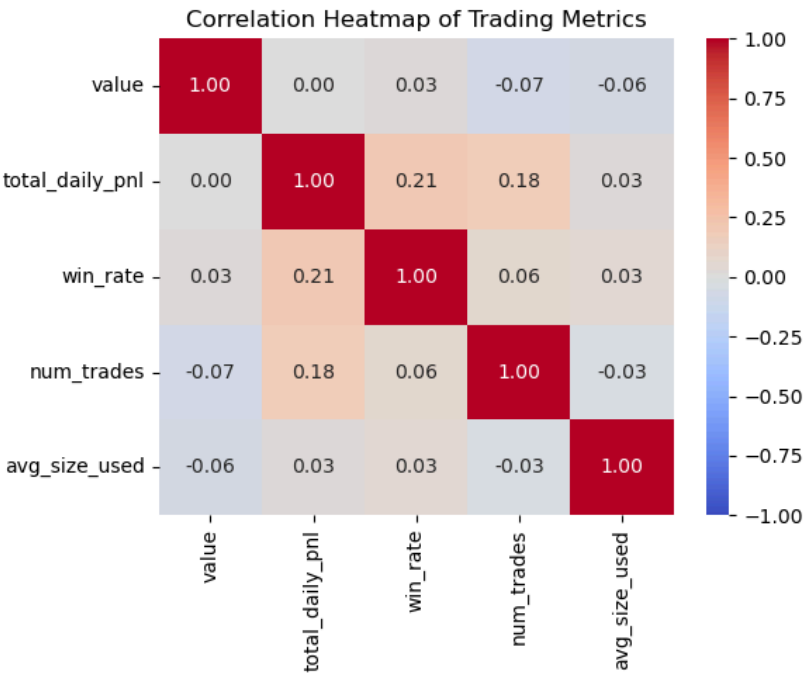
relatively unchanged activity levels, indicating weaker adaptation to market conditions.



3. Sentiment alone does not explain profitability

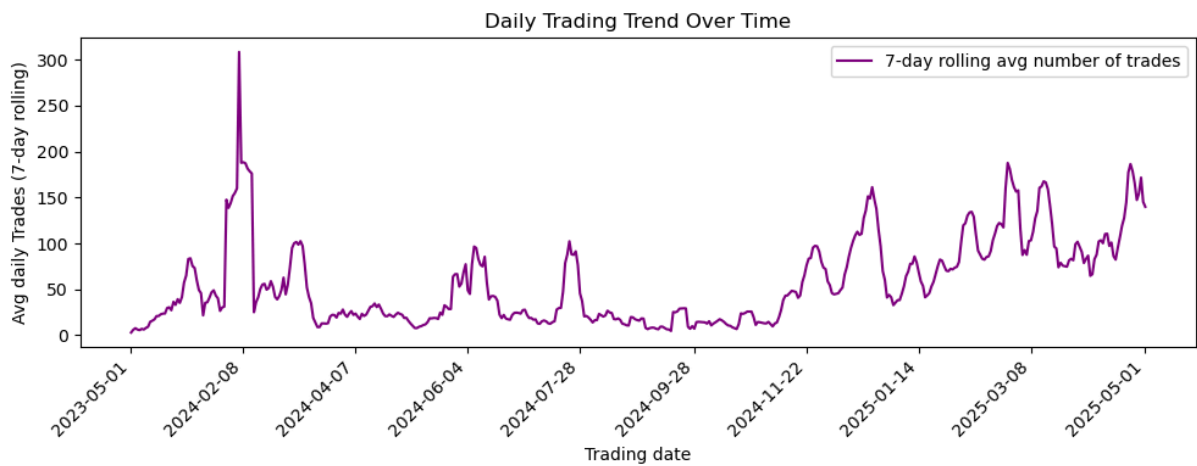
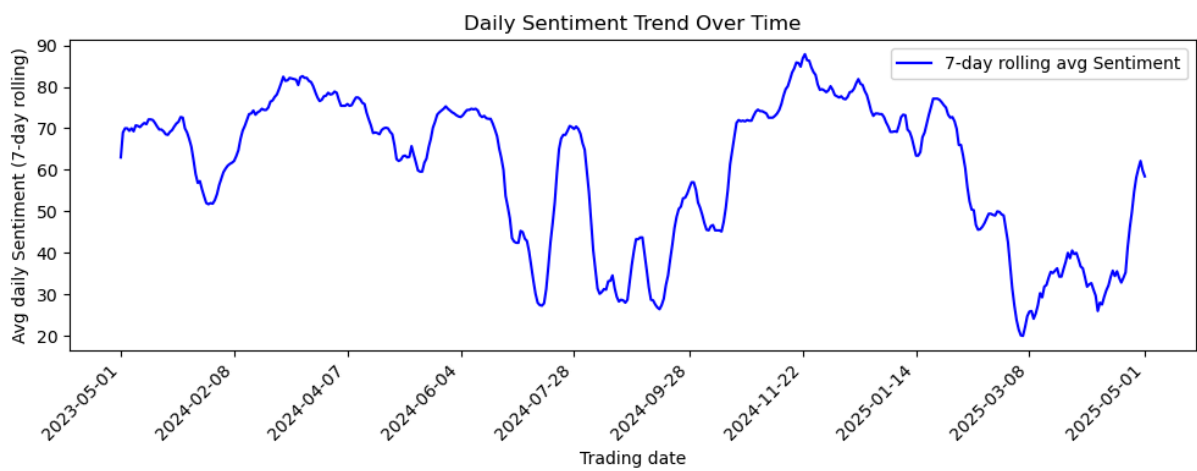
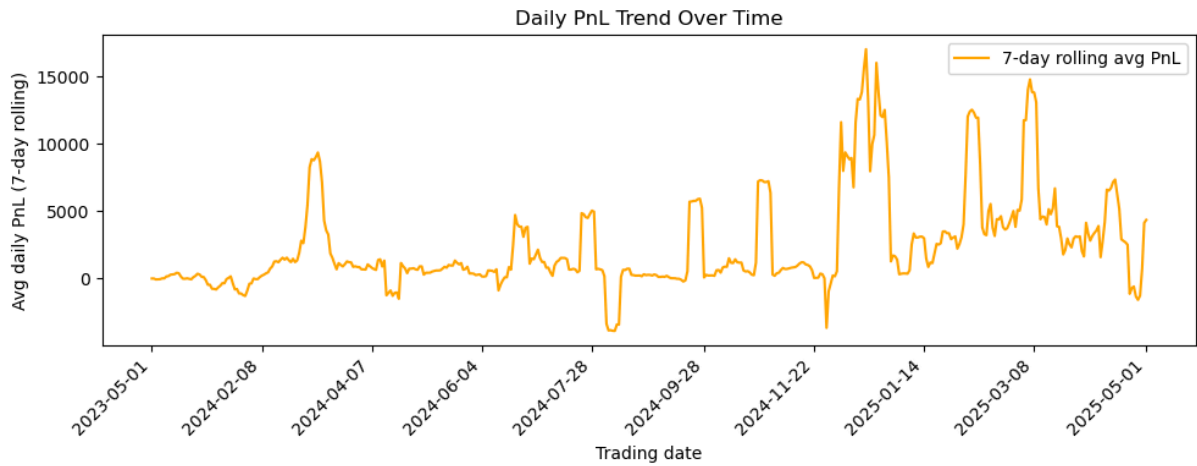
Correlation analysis shows only weak linear relationship between sentiment values and key performance variables such as PnL, number of trades, and win

rate. Heat maps confirm that sentiment by itself is not a strong predictor of trader success, suggesting that individual trading behaviour and strategy play a larger role.



4 Time-series trends reveal regime-dependent profitability

A 7-day rolling average analysis shows that trader PnL remained low and unstable through most of 2023 and mid-2024. From late 2024 into early 2025, profitability increased sharply, coinciding with sustained Greed and Extreme Greed in market sentiment and higher trading activity. When sentiment falls back into strong fear, both trading activity and profitability tend to decline or become more volatile.



5. Conclusion

This analysis suggests that while overall market sentiment influences trader activity and volatility, long-term profitability is driven more by trader skill and adaptive behaviour than by sentiment alone. Top traders demonstrate resilience

across all sentiment regimes, while weaker traders fail to capitalize even during favorable market conditions.

6. Limitations & Next Steps

This study is exploratory and does not attempt to predict PnL. The data covers a limited time period and a single market. Future work could include building regression or machine learning models to quantify the impact of sentiment and trading behaviour on profitability. clustering traders into behavioural types and back-testing sentiment-based trading strategies